Selectors

Pick what to style

Basic

`p { }`	element
`.card { }`	class
`#main { }`	id
`* { }`	everything
`a, b { }`	group
`div p { }`	descendant
`div > p { }`	direct child
`h1 + p { }`	next sibling
`h1 ~ p { }`	all siblings after

Attribute

`[type="text" ]`	exact match
`[href^="https"]`	starts with
`[href$=".pdf"]`	ends with
`[class*="btn"]`	contains

Pseudo

`:hover  :focus  :active`	user states
`:first-child  :last-child`	position
`:nth-child(2n)`	every even
`:not(.x)`	exclude
`::before  ::after`	generated content
`::placeholder  ::selection`	special parts
`:is(h1, h2)  :has(img)`	modern selectors

Cascade & Specificity

Who wins the style fight

Specificity (low to high)

`*`	weakest, zero weight
`p, ::before`	element: 0-0-1
`.card, [type], :hover`	class: 0-1-0
`#main`	id: 1-0-0
`style="color: red"`	inline beats selectors
`color: red !important;`	beats almost all

Inheritance

`inherit`	copy from parent
`initial`	browser default value
`unset`	inherit or initial
`revert`	back to browser style
`all: unset;`	reset every property

Organizing

`@layer base, components;`	order layers
`@layer base { }`	put rules in a layer
`:where(.a)`	zero specificity
`:is(.a, .b)`	takes highest specificity
`@import url("a.css");`	load another file
`/* comment */`	comment

Box & Text

Spacing and type

Box Model

`width: 300px;`	set width
`height: 100px;`	set height
`padding: l0px 20px;`	inside space
`margin: 0 auto;`	outside / center
`border: 1px solid #333;`	border
`border-radius: 8px;`	round corners
`box-sizing: border-box;`	include padding
`overflow: hidden;`	clip extra

Text & Font

`color: #ff0000;`	text color
`font-size: 16px;`	size
`font-weight: 700;`	boldness
`font-family: Arial, sans-serif;`	font stack
`line-height: 1.5;`	line spacing
`text-align: center;`	align text
`text-decoration: underline;`	underline / none
`text-transform: uppercase;`	change case
`letter-spacing: 1px;`	letter gap

Background

`background: #fff;`	color
`background-image: url(a.png);`	image
`background-size: cover;`	fill area
`linear-gradient(red, blue)`	gradient

Colors & Backgrounds

Paint your page

Color Formats

`color: red;`	named color
`color: #3b82f6;`	hex
`color: #3b82f680;`	hex with alpha
`color: rgb(59 130 246);`	rgb
`color: rgb(0 0 0 / 50%);`	rgb with alpha
`color: hsl(210 90% 50%);`	hue, sat, light
`color: currentColor;`	use text color

Gradients

`linear-gradient(90deg, red, blue)`	straight gradient
`radial-gradient(circle, #fff, #000)`	round gradient
`conic-gradient(red, blue, red)`	around a center
`repeating-linear-gradient()`	stripes

Backgrounds

`background-repeat: no-repeat;`	do not tile
`background-position: center;`	place the image
`background-attachment: fixed;`	parallax effect
`background-clip: text;`	paint inside text
`background: url(a.jpg) center/cover;`	shorthand
`opacity: 0.8;`	whole element see-through

Typography & Fonts

Make text look good

Web Fonts

`@font-face { font-family: "X"; }`	load custom font
`src: url(a.woff2) format("woff2");`	font file
`font-display: swap;`	show text at once
`font-family: system-ui, sans-serif;`	system font stack

Text Control

`font-style: italic;`	slanted text
`font-variant: small-caps;`	small capitals
`text-indent: 2em;`	indent first line
`text-shadow: 1px 1px 2px #000;`	text shadow
`word-spacing: 4px;`	gap between words
`white-space: nowrap;`	no wrapping

Long Text

`text-overflow: ellipsis;`	show …
`overflow-wrap: anywhere;`	break long words
`hyphens: auto;`	hyphenate text
`line-clamp: 3;`	limit to 3 lines
`columns: 2;`	newspaper columns
`text-wrap: balance;`	even line lengths

Flexbox

One-dimensional layout

Container

`display: flex;`	start flexbox
`flex-direction: row | column;`	main axis
`flex-wrap: wrap;`	allow wrapping
`justify-content: center;`	main axis align
`align-items: center;`	cross axis align
`gap: 16px;`	space between
`align-content: space-between;`	rows spacing

Justify Values

`flex-start  flex-end`	edges
`center`	middle
`space-between`	edges + even gaps
`space-around  space-evenly`	even spacing

Items

`flex: 1;`	grow to fill
`flex-grow  flex-shrink`	size rules
`flex-basis: 200px;`	start size
`order: 2;`	change order
`align-self: flex-end;`	one item align

Quick Center

`display:flex; place-items:center;`	center anything

Grid Deep Dive

Areas, tracks, alignment

Templates

`grid-template-areas: "h h" "s m";`	name layout areas
`grid-area: m;`	place into area
`grid-template-columns: 200px 1fr;`	fixed + flexible
`grid-auto-rows: 100px;`	size for new rows
`grid-auto-flow: dense;`	fill the gaps
`minmax(100px, 1fr)`	size range

Placement

`grid-column: 2 / 4;`	from line 2 to 4
`grid-column: span 2;`	cover 2 columns
`grid-row: 1 / -1;`	whole height
`order: -1;`	move earlier

Alignment

`justify-items: center;`	items across
`align-items: center;`	items down
`place-content: center;`	whole grid centered
`justify-self: end;`	one item across
`align-self: start;`	one item down
`auto-fill  auto-fit`	fill or collapse tracks

Grid & Position

Layout tools

Grid

`display: grid;`	start grid
`grid-template-columns: 1fr 2fr;`	column sizes
`grid-template-columns: repeat(3, 1fr);`	3 equal columns
`grid-template-rows: auto 1fr;`	row sizes
`gap: 10px;`	grid gap
`grid-column: 1 / 3;`	span columns
`grid-area: header;`	named area
`repeat(auto-fit, minmax(200px, 1fr))`	responsive grid

Position

`position: static;`	default
`position: relative;`	offset in place
`position: absolute;`	vs nearest parent
`position: fixed;`	stay on screen
`position: sticky; top: 0;`	stick on scroll
`top  right  bottom  left`	offsets
`z-index: 10;`	stack order

Display

`display: none;`	hide + remove
`display: block | inline | inline-block`	box type
`visibility: hidden;`	hide, keep space

Transforms & Filters

Move, rotate, blur

Transform

`transform: translate(10px, 20px);`	move
`transform: scale(1.2);`	grow
`transform: rotate(45deg);`	rotate
`transform: skewX(10deg);`	slant
`transform-origin: top left;`	pivot point
`perspective: 800px;`	3D depth
`transform: rotateY(180deg);`	flip in 3D

Filters

`filter: blur(4px);`	blur
`filter: grayscale(100%);`	black and white
`filter: brightness(1.2);`	lighter
`filter: contrast(1.3);`	more contrast
`filter: drop-shadow(0 2px 4px #0005);`	shape shadow
`backdrop-filter: blur(8px);`	frosted glass

Blend & Clip

`mix-blend-mode: multiply;`	blend with below
`clip-path: circle(50%);`	cut to a circle
`object-fit: cover;`	fill image box
`aspect-ratio: 16 / 9;`	fixed ratio

Effects

Motion and responsive

Transitions

`transition: all 0.3s ease;`	smooth change
`transform: scale(1.1);`	resize
`transform: rotate(45deg) ;`	rotate
`transform: translateX(20px);`	move
`box-shadow: 0 4px 10px #0003;`	shadow
`opacity: 0.5;`	transparency
`filter: blur(4px);`	blur / grayscale

Animation

`@keyframes move { to { left: 9px } }`	define steps
`animation: move 2s infinite;`	run animation
`animation-delay: 1s;`	wait first
`cursor: pointer;`	hand cursor

Responsive

`@media (max-width: 768px) { }`	mobile styles
`@media (prefers-color-scheme: dark)`	dark mode
`min-width  max-width`	size limits
`img { max-width: 100%; }`	fluid images

Responsive Design

Fit every screen

Media Queries

`@media (min-width: 768px) { }`	tablet and up
`@media (max-width: 600px) { }`	phones
`@media (orientation: landscape)`	wide screens
`@media print { }`	printing styles
`@media (prefers-reduced-motion)`	less animation
`@media (hover: hover)`	mouse devices

Container Queries

`container-type: inline-size;`	make a container
`@container (min-width: 400px) { }`	style by parent width
`cqw  cqh`	container units

Fluid Tools

`width: min(100%, 600px);`	capped width
`font-size: clamp(1rem, 2vw, 2rem);`	fluid font
`dvh  svh  lvh`	mobile viewport units
`<meta name="viewport" ...>`	needed in HTML
`img { max-width: 100%; }`	fluid image

Forms & States

Style user input

Form States

`:focus-visible`	keyboard focus only
`:focus-within`	child has focus
`:checked`	checked box
`:disabled  :enabled`	input state
`:required  :optional`	required fields
`:valid  :invalid`	validation result

Form Styling

`accent-color: #3b82f6;`	tint checkboxes
`caret-color: red;`	cursor color
`outline: 2px solid blue;`	focus ring
`outline-offset: 2px;`	gap from edge
`appearance: none;`	remove default look
`resize: none;`	lock textarea size

Interaction

`cursor: pointer;`	hand cursor
`user-select: none;`	block selecting
`pointer-events: none;`	ignore clicks
`scroll-behavior: smooth;`	smooth anchor scroll
`scroll-snap-type: x mandatory;`	snap carousel

Modern CSS

Newer features

Nesting & Selectors

`.card { &:hover { } }`	native nesting
`.card { .title { } }`	nested child
`:has(> img)`	parent selector
`:not(:first-child)`	exclude first
`:nth-child(odd)`	every odd item

Logical Properties

`margin-inline: auto;`	left and right
`padding-block: 1rem;`	top and bottom
`inset: 0;`	top/right/bottom/left
`inline-size  block-size`	width / height
`border-start-start-radius`	direction-aware corner

New Tools

`color-mix(in srgb, red, blue)`	mix colors
`@property --x { syntax: "<number>"; }`	typed variable
`@starting-style { }`	animate on entry
`view-transition-name: hero;`	page transitions
`scroll-timeline`	scroll-driven animation
`text-wrap: pretty;`	nicer line breaks

Layout Recipes

Copy and adapt

Centering

`margin: 0 auto;`	center a block
`display: grid; place-items: center;`	center anything
`position: absolute; inset: 0; margin: auto;`	center overlay

Common Layouts

`.wrap { max-width: 1100px; margin: auto; }`	page container
`repeat(auto-fit, minmax(240px, 1fr))`	responsive card grid
`display: flex; gap: 1rem;`	simple row
`flex: 1 1 0;`	equal columns

Sticky & Fullscreen

`min-height: 100dvh;`	full screen height
`position: sticky; top: 0;`	sticky header
`grid-template-rows: auto 1fr auto;`	sticky footer layout
`overflow: auto;`	scroll inside box
`overscroll-behavior: contain;`	stop scroll chaining

Tips & Units

Variables, units, habits

Variables

`:root { --main: #3b82f6; }`	define variable
`color: var(--main);`	use variable
`var(--x, red)`	with fallback
`calc(100% - 20px)`	do math
`clamp(1rem, 2vw, 2rem)`	fluid size

Units

`px`	fixed pixels
`%`	parent based
`em  rem`	font based
`vw  vh`	viewport size
`fr`	grid fraction

Good Habits

`* { box-sizing: border-box; }`	easy sizing
`Mobile first design`	small screens first
`Use classes, not ids`	easy reuse

Keep coding. Keep learning.

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